

Highlights

- Knowledge-based system couples hierarchical feature encoding with multi-method XAI methodology
- SHAP, permutation, and ALE converge; LIME diverges under correlated survey features
- Tuned XGBoost inference engine: AUC=0.903 for classification, RMSE=54.10 for regression (613K students)
- SES disparities (ABROCA=0.023) are the largest fairness concern across 80 countries
- Reproducible single-workstation pipeline; cross-context generalization limits transparently documented